七 多元函数微分学

一、选择题

1. (**D**). 由$\frac{\partial z}{\partial x}>0$及$x_1<x_2$,得$f(x_1,y)<f(x_2,y)$,从而$f(x_1,y_1)<f(x_2,y_1)$.

由$\frac{\partial z}{\partial y}<0$及$y_1>y_2$,得$f(x,y_1)<f(x,y_2)$,从而$f(x_2,y_1)<f(x_2,y_2)$.

故$f(x_1,y_1)<f(x_2,y_2)$. 选(D).

2. (**A**). 因为$\lim\limits_{\substack{x\to 0\\ y=0}} f(x,y)=\lim\limits_{x\to 0}\frac{x}{|x|}$不存在,所以$\lim\limits_{\substack{x\to 0\\ y\to 0}} f(x,y)$不存在,故$f(x,y)$在(0,0)处不连续.

因为$\lim\limits_{x\to 0}\frac{f(x,0)-f(0,0)}{x}=\lim\limits_{x\to 0}\frac{1}{|x|}=+\infty$,所以$f(x,y)$在(0,0)处对$x$不可偏导,同理,$f(x,y)$在(0,0)处对$y$也不可偏导. 选(A).

3. (**C**). 因为$\lim\limits_{\substack{y\to 0\\ x=y^2}} f(x,y)=\frac{1}{2}\neq\lim\limits_{\substack{y\to 0\\ x=-y^2}} f(x,y)=-\frac{1}{2}$,所以$f(x,y)$在(0,0)处不连续.

由$\lim\limits_{x\to 0}\frac{f(x,0)-f(0,0)}{x}=\lim\limits_{x\to 0}\frac{0}{x^3}=0$,$\lim\limits_{y\to 0}\frac{f(0,y)-f(0,0)}{y}=\lim\limits_{y\to 0}\frac{0}{y^5}=0$,得$f'_x(0,0)=0$,$f'_y(0,0)=0$,即$f(x,y)$在(0,0)处可偏导. 选(C).

4. (**C**). $\lim\limits_{x\to 0}\frac{f(x,0)-f(0,0)}{x}=\lim\limits_{x\to 0}\frac{e^{|x|}-1}{x}$不存在,则$f(x,y)$在(0,0)处对$x$不可偏导.

由$\lim\limits_{y\to 0}\frac{f(0,y)-f(0,0)}{y}=\lim\limits_{y\to 0}\frac{e^{y^2}-1}{y}=0$,得$f'_y(0,0)=0$,则$f(x,y)$在(0,0)处对$y$可偏导. 选(C).

5. (**D**). 由$\lim\limits_{\substack{x\to 0\\ y\to 0}} f(x,y)=f(0,0)=0$,得$f(x,y)$在(0,0)处连续.

由$\lim\limits_{x\to 0}\frac{f(x,0)-f(0,0)}{x}=0$,得$f'_x(0,0)=0$.

由$\lim\limits_{y\to 0}\frac{f(0,y)-f(0,0)}{y}=\lim\limits_{y\to 0}\arctan\frac{1}{|y|}=\frac{\pi}{2}$,得$f'_y(0,0)=\frac{\pi}{2}$,即$f(x,y)$在(0,0)处可偏导.

令$\rho=\sqrt{x^2+y^2}$,

$$\lim_{\rho\to 0}\frac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\rho}=\lim_{\rho\to 0} y\cdot\frac{\arctan\frac{1}{\rho}-\frac{\pi}{2}}{\rho},$$

而$\lim\limits_{\rho\to 0}\frac{\arctan\frac{1}{\rho}-\frac{\pi}{2}}{\rho}=\lim\limits_{\rho\to 0}\frac{1}{1+\frac{1}{\rho^2}}\cdot\left(-\frac{1}{\rho^2}\right)=-\lim\limits_{\rho\to 0}\frac{1}{1+\rho^2}=-1$.

故$\lim\limits_{\rho\to 0}\frac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\rho}=0$,即$f(x,y)$在(0,0)处可微. 选(D).

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6.(B). 令 $\rho=\sqrt{(x-1)^2+(y+1)^2}$，由 $\lim\limits_{\substack{x\to1\\y\to-1}}\dfrac{f(x,y)-2x-y+2}{(x-1)^2+(y+1)^2}=2$，得

$$f(1,-1)=-1,f(x,y)-2x-y+2=o(\rho),$$

即 $\Delta u=f(x,y)-f(1,-1)=2(x-1)+(y+1)+o(\rho)$，则 $f(x,y)$ 在点 $(1,-1)$ 处可微，且 $\mathrm{d}u|_{(1,-1)}=2\mathrm{d}x+\mathrm{d}y$，从而 $f'_x(1,-1)=2,f'_y(1,-1)=1$.

$$\frac{\mathrm{d}z}{\mathrm{d}t}=2tf'_x[t^2,f(t^2,t^3)]+f'_y[t^2,f(t^2,t^3)]\cdot[2tf'_x(t^2,t^3)+3t^2f'_y(t^2,t^3)],$$

故 $\left.\dfrac{\mathrm{d}z}{\mathrm{d}t}\right|_{t=-1}=-2f'_x(1,-1)+f'_y(1,-1)\cdot[-2f'_x(1,-1)+3f'_y(1,-1)]=-5$. 选(B).

7.(D). $\dfrac{\partial u}{\partial x}=2xf'_1+yf'_2+2xg-yg'$，

$$\frac{\partial^2u}{\partial x\partial y}=2x(2yf''_{11}+xf''_{12})+f'_2+y(2yf''_{21}+xf''_{22})+2g'-g'-\frac{y}{x}g''$$

$$=4xyf''_{11}+2(x^2+y^2)f''_{12}+xyf''_{22}+f'_2+g'-\frac{y}{x}g''.$$

选(D).

8.(A). 将 $x=0,y=1$ 代入，得 $\mathrm{e}^z+z=1$，解得 $z=0$.

$\mathrm{e}^{\cos xy+y^2+z-2}+z=2x+y$ 两边分别同时对 x,y 求偏导，得

$$\mathrm{e}^{\cos xy+y^2+z-2}\cdot\left(-y\sin xy+\frac{\partial z}{\partial x}\right)+\frac{\partial z}{\partial x}=2,\mathrm{e}^{\cos xy+y^2+z-2}\cdot\left(-x\sin xy+2y+\frac{\partial z}{\partial y}\right)+\frac{\partial z}{\partial y}=1,$$

将 $x=0,y=1,z=0$ 代入，得 $\left.\dfrac{\partial z}{\partial x}\right|_{(0,1)}=1,\left.\dfrac{\partial z}{\partial y}\right|_{(0,1)}=-\dfrac{1}{2}$. 选(A).

9.(B). 由 $\lim\limits_{\substack{x\to1\\y\to1}}\dfrac{f(x,y)-3}{\sin^2\pi x+\sin^2\pi y}=2$，得 $f(1,1)=3$，因为 $\lim\limits_{\substack{x\to1\\y\to1}}\dfrac{f(x,y)-3}{\sin^2\pi x+\sin^2\pi y}=2>0$，由极限的局部保号性可知，存在 $\delta>0$，当 $0<\sqrt{(x-1)^2+(y-1)^2}<\delta$ 时，$\dfrac{f(x,y)-3}{\sin^2\pi x+\sin^2\pi y}>0$，而 $\sin^2\pi x+\sin^2\pi y>0$，所以 $f(x,y)>3=f(1,1)$，即 $f(1,1)$ 是极小值. 选(B).

10.(C). 由 $\begin{cases}\dfrac{\partial z}{\partial x}=\mathrm{e}^{2x}(2x+2y^2+4y+1)=0,\\[2mm]\dfrac{\partial z}{\partial y}=\mathrm{e}^{2x}(2y+2)=0,\end{cases}$ 得 $\begin{cases}x=\dfrac{1}{2},\\y=-1.\end{cases}$

$\dfrac{\partial^2z}{\partial x^2}=4\mathrm{e}^{2x}(x+y^2+2y+1),\dfrac{\partial^2z}{\partial x\partial y}=4\mathrm{e}^{2x}(y+1),\dfrac{\partial^2z}{\partial y^2}=2\mathrm{e}^{2x}$，则

$$A=\left.\frac{\partial^2z}{\partial x^2}\right|_{(\frac12,-1)}=2\mathrm{e},B=\left.\frac{\partial^2z}{\partial x\partial y}\right|_{(\frac12,-1)}=0,C=\left.\frac{\partial^2z}{\partial y^2}\right|_{(\frac12,-1)}=2\mathrm{e},$$

因为 $AC-B^2>0$ 且 $A>0$，所以 $\left(\dfrac{1}{2},-1\right)$ 为极小值点，且极小值为 $f\left(\dfrac{1}{2},-1\right)=-\dfrac{\mathrm{e}}{2}$. 选(C).

11.(B). 由 $\begin{cases}\dfrac{\partial z}{\partial x}=\mathrm{e}^{2x}(2x^2+2x+2y^2+4y-2)=0,\\[2mm]\dfrac{\partial z}{\partial y}=\mathrm{e}^{2x}(2y+2)=0,\end{cases}$ 得 $\begin{cases}x=-2,\\y=-1\end{cases}$ 或 $\begin{cases}x=1,\\y=-1,\end{cases}$

$$\frac{\partial^2z}{\partial x^2}=\mathrm{e}^{2x}(4x^2+8x+4y^2+8y-2),\frac{\partial^2z}{\partial x\partial y}=\mathrm{e}^{2x}(4y+4),\frac{\partial^2z}{\partial y^2}=2\mathrm{e}^{2x}.$$

当 $\begin{cases}x=-2,\\y=-1\end{cases}$ 时，$A=-6\mathrm{e}^{-4},B=0,C=2\mathrm{e}^{-4}$，因为 $AC-B^2<0$，所以 $(-2,-1)$ 不是极值点.

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当$\begin{cases}x=1,\\y=-1\end{cases}$时，$A=6e^2$，$B=0$，$C=2e^2$，因为$AC-B^2>0$且$A>0$，所以$(1,-1)$是极小值点. 选(B).

12. **(B)**. 设(x_0,y_0)为$f(x,y)$在区域D内的驻点，由$A+C=0$，$B\neq 0$，得$AC-B^2<0$，则(x_0,y_0)不是$f(x,y)$的极值点，即$f(x,y)$在区域D内没有极值点，也取不到最值. 因为$f(x,y)$在区域D上连续，所以$f(x,y)$在区域D上可以取到最小值和最大值，故$f(x,y)$在区域D的边界上取到最小值和最大值. 选(B).

13. **(A)**. $\dfrac{\partial z}{\partial x}=2x\sin\dfrac{\pi}{4}y$，$\dfrac{\partial z}{\partial y}=\dfrac{\pi}{4}x^2\cos\dfrac{\pi}{4}y$，$\left.\dfrac{\partial z}{\partial x}\right|_{(1,1)}=\sqrt{2}$，$\left.\dfrac{\partial z}{\partial y}\right|_{(1,1)}=\dfrac{\sqrt{2}\pi}{8}$.

$\overrightarrow{AB}=\{-1,1\}$，$\cos\alpha=-\dfrac{\sqrt{2}}{2}$，$\cos\beta=\dfrac{\sqrt{2}}{2}$.

则所求的方向导数为$\sqrt{2}\times\left(-\dfrac{\sqrt{2}}{2}\right)+\dfrac{\sqrt{2}\pi}{8}\times\dfrac{\sqrt{2}}{2}=\dfrac{\pi}{8}-1$. 选(A).

二、填空题

1. **解** $\dfrac{\partial z}{\partial x}=\dfrac{1}{1+\left(\dfrac{x+y}{1-xy}\right)^2}\cdot\dfrac{1-xy+y(x+y)}{(1-xy)^2}=\dfrac{1+y^2}{(1+x^2)(1+y^2)}=\dfrac{1}{1+x^2}$.

由对称性得$\dfrac{\partial z}{\partial y}=\dfrac{1}{1+y^2}$，故$\dfrac{\partial z}{\partial x}+\dfrac{\partial z}{\partial y}=\dfrac{1}{1+x^2}+\dfrac{1}{1+y^2}=\dfrac{2+x^2+y^2}{(1+x^2)(1+y^2)}$.

2. **解** $\dfrac{\partial z}{\partial x}=\dfrac{2x}{x^2+y^2}$，$\dfrac{\partial^2 z}{\partial x^2}=2\dfrac{(x^2+y^2)-2x^2}{(x^2+y^2)^2}=\dfrac{2(y^2-x^2)}{(x^2+y^2)^2}$.

由对称性得$\dfrac{\partial^2 z}{\partial y^2}=\dfrac{2(x^2-y^2)}{(x^2+y^2)^2}$，故$\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=0$.

3. **解** 令$\begin{cases}x+y=u,\\\dfrac{y}{x}=v,\end{cases}$解得$x=\dfrac{u}{v+1}$，$y=\dfrac{uv}{v+1}$，代入$f\left(x+y,\dfrac{y}{x}\right)=x^2+y^2$，得

$$f(u,v)=u^2\left[1-\frac{2v}{(v+1)^2}\right]\Rightarrow f(x,y)=x^2\left[1-\frac{2y}{(y+1)^2}\right],$$

$\dfrac{\partial f}{\partial x}=2x\left[1-\dfrac{2y}{(y+1)^2}\right]$，$\dfrac{\partial f}{\partial y}=\dfrac{2x^2(y-1)}{(y+1)^3}$，则$dz=2x\left[1-\dfrac{2y}{(y+1)^2}\right]dx+\dfrac{2x^2(y-1)}{(y+1)^3}dy$.

4. **解**

$$\frac{\partial z}{\partial x}=2f'_1\left(2x-y,\frac{y}{x}\right)-\frac{y}{x^2}f'_2\left(2x-y,\frac{y}{x}\right),$$

$$\frac{\partial z}{\partial y}=-f'_1\left(2x-y,\frac{y}{x}\right)+\frac{1}{x}f'_2\left(2x-y,\frac{y}{x}\right),$$

则$\left.\dfrac{\partial z}{\partial x}\right|_{(1,3)}=2f'_1(-1,3)-3f'_2(-1,3)=-7$，$\left.\dfrac{\partial z}{\partial y}\right|_{(1,3)}=-f'_1(-1,3)+f'_2(-1,3)=3$，

故$\left.dz\right|_{(1,3)}=-7dx+3dy$.

5. **解** 通过计算得$\dfrac{\partial z}{\partial x}=2xf'_1-\dfrac{y}{x^2}f'_2$，

$$\frac{\partial^2 z}{\partial x\partial y}=2x\left(2yf''_{11}+\frac{1}{x}f''_{12}\right)-\frac{1}{x^2}f'_2-\frac{y}{x^2}\left(2yf''_{21}+\frac{1}{x}f''_{22}\right)$$

$$=-\frac{1}{x^2}f'_2+4xyf''_{11}+2\left(1-\frac{y^2}{x^2}\right)f''_{12}-\frac{y}{x^3}f''_{22}.$$

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6. 解 $z=f(x+y,y+z,z+x)$ 两边同时对 x 求偏导，得 $\frac{\partial z}{\partial x}=f_1'+f_2'\frac{\partial z}{\partial x}+f_3'\left(1+\frac{\partial z}{\partial x}\right)$，

解得 $\frac{\partial z}{\partial x}=\frac{f_1'+f_3'}{1-f_2'-f_3'}$.

7. 解 $f_x'(x,y,z)=y\left(\mathrm{e}^x z^2+2z\mathrm{e}^x\frac{\partial z}{\partial x}\right)$，$x+y+z+xyz=0$ 两边同时对 x 求偏导，得

$1+\frac{\partial z}{\partial x}+y\left(z+x\frac{\partial z}{\partial x}\right)=0$，将 $x=0,y=1,z=-1$ 代入，得 $\begin{cases}f_x'(0,1,-1)=1-2\frac{\partial z}{\partial x},\\ 1+\frac{\partial z}{\partial x}-1=0,\end{cases}$

解得 $f_x'(0,1,-1)=1$.

8. 解 将 $x=\frac{1}{2},y=\frac{1}{2}$ 代入 $\mathrm{e}^{2yz}+x+y^2+z=\frac{7}{4}$，得 $z=0$.

$\mathrm{e}^{2yz}+x+y^2+z=\frac{7}{4}$ 两边同时求微分，得 $2\mathrm{e}^{2yz}(z\mathrm{d}y+y\mathrm{d}z)+\mathrm{d}x+2y\mathrm{d}y+\mathrm{d}z=0$.

将 $x=\frac{1}{2},y=\frac{1}{2},z=0$ 代入，得 $\mathrm{d}z\Big|_{\left(\frac{1}{2},\frac{1}{2}\right)}=-\frac{1}{2}\mathrm{d}x-\frac{1}{2}\mathrm{d}y$.

9. 解 令 $\rho=\sqrt{(x-1)^2+y^2}$，由 $\lim\limits_{\substack{x\to1\\y\to0}}\frac{f(x,y)-3x+y+2}{\sqrt{(x-1)^2+y^2}}=0$，得

$$f(1,0)=1,f(x,y)-3x+y+2=o(\rho),$$

则 $\Delta z=f(x,y)-f(1,0)=3(x-1)-y+o(\rho)$，即

$$\mathrm{d}z|_{(1,0)}=3\mathrm{d}x-\mathrm{d}y,f_x'(1,0)=3,f_y'(1,0)=-1,$$

故 $\lim\limits_{h\to0}\frac{f(\mathrm{e}^{2h},h)-f(1,-h)}{\ln(1+2h)}=\frac{1}{2}\lim\limits_{h\to0}\frac{f(\mathrm{e}^{2h},h)-f(1,-h)}{h}$，其中

$$\begin{aligned}f(\mathrm{e}^{2h},h)-f(1,-h)&=[f(\mathrm{e}^{2h},h)-f(1,h)]+[f(1,h)-f(1,-h)]\\&=f_x'(\xi,h)(\mathrm{e}^{2h}-1)+2hf_y'(1,\eta),\end{aligned}$$

ξ 介于 1 与 e^{2h} 之间，η 介于 $-h$ 与 h 之间，故

$$\text{原式}=\frac{1}{2}\left[\lim_{h\to0}f_x'(\xi,h)\cdot\frac{\mathrm{e}^{2h}-1}{h}+2\lim_{h\to0}f_y'(1,\eta)\right]=\frac{1}{2}[2f_x'(1,0)+2f_y'(1,0)]=2.$$

10. 解 由 $\frac{\partial Q}{\partial x}=2x$，得 $Q(x,y)=x^2+\varphi(y)$.

再由 $Q(1,y)=\sin2y-y$，得 $Q(x,y)=x^2+\sin2y-y-1$.

11. 解 $\frac{\partial z}{\partial x}=2x\mathrm{e}^{2y},\frac{\partial z}{\partial y}=2x^2\mathrm{e}^{2y}$，则 $\frac{\partial z}{\partial x}\Big|_{(1,1)}=2\mathrm{e}^2,\frac{\partial z}{\partial y}\Big|_{(1,1)}=2\mathrm{e}^2$.

又由于 $\overrightarrow{AB}=\{-1,1\}$，$\cos\alpha=-\frac{1}{\sqrt{2}},\cos\beta=\frac{1}{\sqrt{2}}$，则所求的方向导数为

$$2\mathrm{e}^2\cdot\left(-\frac{1}{\sqrt{2}}\right)+2\mathrm{e}^2\cdot\frac{1}{\sqrt{2}}=0.$$

12. 解

$$\frac{\partial u}{\partial x}=2x,\frac{\partial u}{\partial y}=\sin\frac{\pi}{2}z,\frac{\partial u}{\partial z}=\frac{\pi}{2}y\cos\frac{\pi}{2}z,$$

$$\frac{\partial u}{\partial x}\Big|_{(1,-1,1)}=2,\frac{\partial u}{\partial y}\Big|_{(1,-1,1)}=1,\frac{\partial u}{\partial z}\Big|_{(1,-1,1)}=0,$$

$u=x^2+y\sin\frac{\pi}{2}z$ 在点 $(1,-1,1)$ 处增长速度最快的方向的方向导数即梯度的长度，故所求值为 $|\mathbf{grad}\,u|_{(1,-1,1)}=\sqrt{5}$.

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三、解答题

1. 解 因为$\lim\limits_{\substack{x\to0\\y\to0}}f(x,y)=\lim\limits_{\substack{x\to0\\y\to0}}|x-y|g(x,y)=0=f(0,0)$，所以$f(x,y)$在$(0,0)$处连续.

$$\lim_{x\to0}\frac{f(x,0)-f(0,0)}{x}=\lim_{x\to0}\frac{f(x,0)}{x}=\lim_{x\to0}\frac{|x|}{x}g(x,0)=g(0,0)\lim_{x\to0}\frac{|x|}{x},$$

因为$g(0,0)\neq0$，所以$\lim\limits_{x\to0}\dfrac{f(x,0)-f(0,0)}{x}$不存在，即$f(x,y)$在$(0,0)$处对$x$不可偏导.

同理，$f(x,y)$在$(0,0)$处对y也不可偏导. 故$f(x,y)$在$(0,0)$处不可微.

2. 解 (1) 由$\dfrac{\partial z}{\partial x}=2xyf'(x^2-y^2)$，$\dfrac{\partial z}{\partial y}=f(x^2-y^2)-2y^2f'(x^2-y^2)$，得

$$\frac{1}{x}\cdot\frac{\partial z}{\partial x}+\frac{1}{y}\cdot\frac{\partial z}{\partial y}=2yf'(x^2-y^2)+\frac{1}{y}f(x^2-y^2)-2yf'(x^2-y^2)=\frac{1}{y}f(x^2-y^2)=\frac{z}{y^2}.$$

(2) $\dfrac{\partial u}{\partial x}=f'_1+2xf'_2$，$\dfrac{\partial u}{\partial y}=f'_1+2yf'_2$；

$$\frac{\partial^2u}{\partial x^2}=f''_{11}+2xf''_{12}+2f'_2+2x(f''_{21}+2xf''_{22})=f''_{11}+4xf''_{12}+4x^2f''_{22}+2f'_2;$$

$$\frac{\partial^2u}{\partial y^2}=f''_{11}+2yf''_{12}+2f'_2+2y(f''_{21}+2yf''_{22})=f''_{11}+4yf''_{12}+4y^2f''_{22}+2f'_2.$$

则$\dfrac{\partial^2u}{\partial x^2}+\dfrac{\partial^2u}{\partial y^2}=2f''_{11}+4(x+y)f''_{12}+4(x^2+y^2)f''_{22}+4f'_2$.

3. 解 $\dfrac{\mathrm{d}z}{\mathrm{d}t}=\mathrm{e}^t(\sin t+\cos t)f'_1+f'_2\sec^2t$.

4. 解 (1) 由$\dfrac{\partial z}{\partial x}=2f'(2x-y)+g'_1(x,xy)+yg'_2(x,xy)$，得

$$\frac{\partial^2z}{\partial x\partial y}=-2f''(2x-y)+xg''_{12}(x,xy)+g'_2(x,xy)+xyg''_{22}(x,xy).$$

(2) 由$\dfrac{\partial z}{\partial x}=2xf'_1+yf'_2+f'_3$，得

$$\begin{aligned}\frac{\partial^2z}{\partial x\partial y}&=2x(2yf''_{11}+xf''_{12})+f'_2+y(2yf''_{21}+xf''_{22})+2yf''_{31}+xf''_{32}\\&=4xyf''_{11}+2(x^2+y^2)f''_{12}+f'_2+xyf''_{22}+2yf''_{31}+xf''_{32}.\end{aligned}$$

5. 证明 $\dfrac{\partial u}{\partial x}=f'(z)\cdot\dfrac{\partial z}{\partial x}$，$z=y+x\varphi(z)$两边同时对$x$求偏导，得

$$\frac{\partial z}{\partial x}=\varphi(z)+x\varphi'(z)\frac{\partial z}{\partial x}\Rightarrow\frac{\partial z}{\partial x}=\frac{\varphi(z)}{1-x\varphi'(z)}\Rightarrow\frac{\partial u}{\partial x}=\frac{\varphi(z)f'(z)}{1-x\varphi'(z)}.$$

$\dfrac{\partial u}{\partial y}=f'(z)\cdot\dfrac{\partial z}{\partial y}$，$z=y+x\varphi(z)$两边同时对$y$求偏导，得

$$\frac{\partial z}{\partial y}=1+x\varphi'(z)\cdot\frac{\partial z}{\partial y}\Rightarrow\frac{\partial z}{\partial y}=\frac{1}{1-x\varphi'(z)}\Rightarrow\frac{\partial u}{\partial y}=\frac{f'(z)}{1-x\varphi'(z)}.$$

所以$\dfrac{\partial u}{\partial x}=\varphi(z)\dfrac{\partial u}{\partial y}$.

6. 证明 $F\left(x+\dfrac{z}{y},y+\dfrac{z}{x}\right)=0$两边同时对$x$求偏导，得

$$\left(1+\frac{1}{y}\cdot\frac{\partial z}{\partial x}\right)F'_1+\frac{x\dfrac{\partial z}{\partial x}-z}{x^2}F'_2=0\Rightarrow\frac{\partial z}{\partial x}=\frac{\dfrac{z}{x^2}F'_2-F'_1}{\dfrac{F'_1}{y}+\dfrac{F'_2}{x}}.$$

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$F\left(x+\frac{z}{y},y+\frac{z}{x}\right)=0$ 两边同时对 y 求偏导，得 $\frac{y\frac{\partial z}{\partial y}-z}{y^2}F'_1+\left(1+\frac{1}{x}\frac{\partial z}{\partial y}\right)F'_2=0\Rightarrow\frac{\partial z}{\partial y}=\frac{\frac{z}{y^2}F'_1-F'_2}{\frac{F'_1}{y}+\frac{F'_2}{x}}$. 于是 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=\frac{\frac{z}{x}F'_2-xF'_1+\frac{z}{y}F'_1-yF'_2}{\frac{F'_1}{y}+\frac{F'_2}{x}}=z-xy$.

7. 解 $z=f[x-y+g(x-y-z)]$ 两边同时对 x 求偏导，得

$$\frac{\partial z}{\partial x}=f'\cdot\left[1+g'\cdot\left(1-\frac{\partial z}{\partial x}\right)\right]\Rightarrow\frac{\partial z}{\partial x}=\frac{f'(1+g')}{1+f'g'}.$$

$z=f[x-y+g(x-y-z)]$ 两边同时对 y 求偏导，得

$$\frac{\partial z}{\partial y}=f'\cdot\left[-1+g'\cdot\left(-1-\frac{\partial z}{\partial y}\right)\right]\Rightarrow\frac{\partial z}{\partial y}=-\frac{f'(1+g')}{1+f'g'}.$$

8. 解 $z=xf(x+y)$ 及 $F(x,y,z)=0$ 两边同时对 x 求导数，得

$$\begin{cases}\frac{\mathrm{d}z}{\mathrm{d}x}=f(x+y)+xf'(x+y)\left(1+\frac{\mathrm{d}y}{\mathrm{d}x}\right),\\ F'_x+F'_y\frac{\mathrm{d}y}{\mathrm{d}x}+F'_z\frac{\mathrm{d}z}{\mathrm{d}x}=0,\end{cases}$$

解得 $\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{(f+xf')F'_z+F'_x}{xf'F'_z+F'_y}$，$\frac{\mathrm{d}z}{\mathrm{d}x}=\frac{(f+xf')F'_y-xf'F'_x}{xf'F'_z+F'_y}$.

9. 解 $y=f(x,t)$ 与 $G(x,y,t)=0$ 两边同时对 x 求导，得

$$\begin{cases}\frac{\mathrm{d}y}{\mathrm{d}x}=f'_1+f'_2\frac{\mathrm{d}t}{\mathrm{d}x},\\ G'_1+G'_2\frac{\mathrm{d}y}{\mathrm{d}x}+G'_3\frac{\mathrm{d}t}{\mathrm{d}x}=0\end{cases}\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{f'_1G'_3-f'_2G'_1}{G'_3+f'_2G'_2}.$$

10. 解 令 $\begin{cases}x+y=u,\\ x-y=v,\end{cases}$ 则 $\begin{cases}x=\frac{1}{2}(u+v),\\ y=\frac{1}{2}(u-v),\end{cases}$ 代入得

$$f(u,v)=\frac{(u+v)^2-(u-v)^2-(u+v)(u-v)}{4}=uv-\frac{1}{4}u^2+\frac{1}{4}v^2,$$

从而 $z=f(x,y)=xy-\frac{1}{4}x^2+\frac{1}{4}y^2$.

由 $\frac{\partial z}{\partial x}=y-\frac{1}{2}x$，$\frac{\partial z}{\partial y}=x+\frac{1}{2}y$，得 $\mathrm{d}z=\left(y-\frac{1}{2}x\right)\mathrm{d}x+\left(x+\frac{1}{2}y\right)\mathrm{d}y$.

11. 解 $\frac{\partial z}{\partial x}=\frac{\partial u}{\partial x}\cdot\frac{\partial z}{\partial u}+\frac{\partial v}{\partial x}\cdot\frac{\partial z}{\partial v}=\frac{\partial z}{\partial u}+\frac{\partial z}{\partial v}$，$\frac{\partial z}{\partial y}=\frac{\partial u}{\partial y}\cdot\frac{\partial z}{\partial u}+\frac{\partial v}{\partial y}\cdot\frac{\partial z}{\partial v}=a\frac{\partial z}{\partial u}+b\frac{\partial z}{\partial v}$；

$\frac{\partial^2 z}{\partial x^2}=\frac{\partial u}{\partial x}\cdot\frac{\partial^2 z}{\partial u^2}+\frac{\partial v}{\partial x}\cdot\frac{\partial^2 z}{\partial u\partial v}+\frac{\partial u}{\partial x}\cdot\frac{\partial^2 z}{\partial v\partial u}+\frac{\partial v}{\partial x}\cdot\frac{\partial^2 z}{\partial v^2}=\frac{\partial^2 z}{\partial u^2}+2\frac{\partial^2 z}{\partial u\partial v}+\frac{\partial^2 z}{\partial v^2}$；

$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial u}{\partial y}\cdot\frac{\partial^2 z}{\partial u^2}+\frac{\partial v}{\partial y}\cdot\frac{\partial^2 z}{\partial u\partial v}+\frac{\partial u}{\partial y}\cdot\frac{\partial^2 z}{\partial v\partial u}+\frac{\partial v}{\partial y}\cdot\frac{\partial^2 z}{\partial v^2}=a\frac{\partial^2 z}{\partial u^2}+(a+b)\frac{\partial^2 z}{\partial u\partial v}+b\frac{\partial^2 z}{\partial v^2}$；

$\frac{\partial^2 z}{\partial y^2}=a\left(\frac{\partial u}{\partial y}\cdot\frac{\partial^2 z}{\partial u^2}+\frac{\partial v}{\partial y}\cdot\frac{\partial^2 z}{\partial u\partial v}\right)+b\left(\frac{\partial u}{\partial y}\cdot\frac{\partial^2 z}{\partial v\partial u}+\frac{\partial v}{\partial y}\cdot\frac{\partial^2 z}{\partial v^2}\right)$

$=a^2\frac{\partial^2 z}{\partial u^2}+2ab\frac{\partial^2 z}{\partial u\partial v}+b^2\frac{\partial^2 z}{\partial v^2}$.

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代入得

$$(3a^2+4a+1)\frac{\partial^2 z}{\partial u^2}+(6ab+4a+4b+2)\frac{\partial^2 z}{\partial u\partial v}+(3b^2+4b+1)\frac{\partial^2 z}{\partial v^2}=0,$$

则 $\begin{cases}3a^2+4a+1=0,\\3b^2+4b+1=0,\\3ab+2a+2b+1\neq 0,\end{cases}$ 解得 $\begin{cases}a=-1,\\b=-\frac{1}{3}\end{cases}$ 或 $\begin{cases}a=-\frac{1}{3},\\b=-1.\end{cases}$

12. 解 由 $f''_{xy}(x,y)=2x\mathrm{e}^y$，得 $f'_x(x,y)=2x\mathrm{e}^y+\varphi(x)$，再由 $f'_x(0,y)=2$，得 $\varphi(x)=2$，则 $f'_x(x,y)=2x\mathrm{e}^y+2$.

由 $f'_x(x,y)=2x\mathrm{e}^y+2$，得 $f(x,y)=x^2\mathrm{e}^y+2x+\psi(y)$，再由 $f(0,y)=y^2$，得 $\psi(y)=y^2$，故 $f(x,y)=x^2\mathrm{e}^y+2x+y^2$.

13. 解 由 $f''_{xy}(x,y)=x\mathrm{e}^y+xy$，得 $f'_x(x,y)=x\mathrm{e}^y+\frac{1}{2}xy^2+\varphi(x)$.

由 $f'_x(x,0)=\cos 2x$，得 $\varphi(x)=\cos 2x-x$，即 $f'_x(x,y)=x\mathrm{e}^y+\frac{1}{2}xy^2+\cos 2x-x$.

由 $f'_x(x,y)=x\mathrm{e}^y+\frac{1}{2}xy^2+\cos 2x-x$，得

$$f(x,y)=\frac{1}{2}x^2\mathrm{e}^y+\frac{1}{4}x^2y^2+\frac{1}{2}\sin 2x-\frac{1}{2}x^2+\varphi(y).$$

由 $f(0,y)=y^2$，得 $\varphi(y)=y^2$. 故 $f(x,y)=\frac{1}{2}x^2\mathrm{e}^y+\frac{1}{4}x^2y^2+\frac{1}{2}\sin 2x-\frac{1}{2}x^2+y^2$.

14. 解 $f(x,y)$ 的定义域为 $D=\{(x,y)\mid y>0\}$.

令 $z=f(x,y)$，由 $\begin{cases}\frac{\partial z}{\partial x}=2x(2+y^2)=0,\\\frac{\partial z}{\partial y}=2x^2y+\ln y+1=0,\end{cases}$ 得 $(x,y)=\left(0,\frac{1}{\mathrm{e}}\right)$，经过计算得

$$\frac{\partial^2 z}{\partial x^2}=2(2+y^2),\quad \frac{\partial^2 z}{\partial x\partial y}=4xy,\quad \frac{\partial^2 z}{\partial y^2}=2x^2+\frac{1}{y},$$

则 $A=\left.\frac{\partial^2 z}{\partial x^2}\right|_{\left(0,\frac{1}{\mathrm{e}}\right)}=2\left(2+\frac{1}{\mathrm{e}^2}\right)$，$B=\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\left(0,\frac{1}{\mathrm{e}}\right)}=0$，$C=\left.\frac{\partial^2 z}{\partial y^2}\right|_{\left(0,\frac{1}{\mathrm{e}}\right)}=\mathrm{e}$.

因为 $AC-B^2>0$ 且 $A>0$，所以 $\left(0,\frac{1}{\mathrm{e}}\right)$ 为 $f(x,y)$ 的极小值点，极小值为 $f\left(0,\frac{1}{\mathrm{e}}\right)=-\frac{1}{\mathrm{e}}$.

15. 解 (1) 由 $\mathrm{d}z=2x\mathrm{d}x-4y\mathrm{d}y$，得 $\mathrm{d}z=\mathrm{d}(x^2-2y^2)$，从而 $f(x,y)=x^2-2y^2+C$.

再由 $f(0,0)=5$，得 $C=5$，故 $f(x,y)=x^2-2y^2+5$.

(2) 当 $x^2+4y^2<4$ 时，由 $\begin{cases}f'_x=2x=0,\\f'_y=-4y=0,\end{cases}$ 得 $\begin{cases}x=0,\\y=0,\end{cases}$ 即 $f(0,0)=5$.

当 $x^2+4y^2=4$ 时，令 $\begin{cases}x=2\cos t,\\y=\sin t\end{cases}$ $(0\leqslant t\leqslant 2\pi)$，则

$$f=4\cos^2 t-2\sin^2 t+5=6\cos^2 t+3.$$

当 $\cos t=0$ 时，$f_{\min}=3$；当 $\cos t=\pm 1$ 时，$f_{\max}=9$.

故 $f(x,y)$ 在区域 D 上的最小值为 $m=3$，最大值为 $M=9$.

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